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DISTRIBUTION UTILITY MEET DUM 2026

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TATA POWER-DDL

BSES Rajdhani Power Limited

NPCL

BSES Yamuna Power Limited

DUM 2026 CONFERENCE SESSION

BESS Applications in Power Distribution - Select Case Studies

Presented by:

Alekhya Datta, Director, TERI

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



teri | THE ENERGY AND RESOURCES INSTITUTE

Creating Innovative Solutions for a Sustainable Future

India’s Energy Transition – Scale & Speed

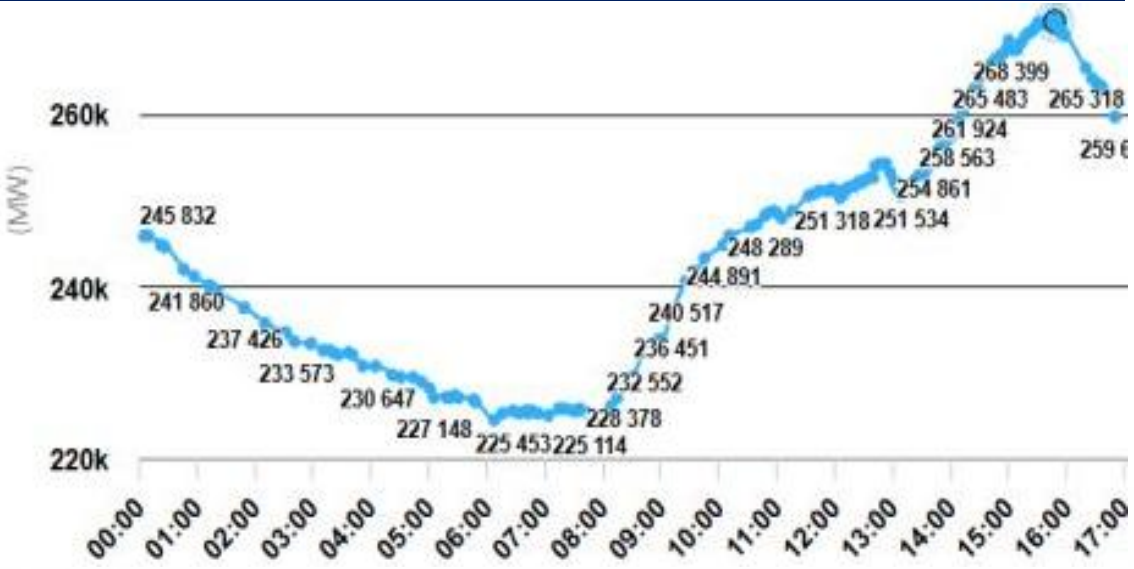
560.5 GW
Total installed capacity (Oct 2026)

54.9%
Non-fossil share of installed capacity

55.3 GW
Non-fossil capacity added in FY2025–26

$350 B
Investment needed to reach 500 GW

Maximum All India Demand Met on 21 May’26 (Thursday) at 15:47

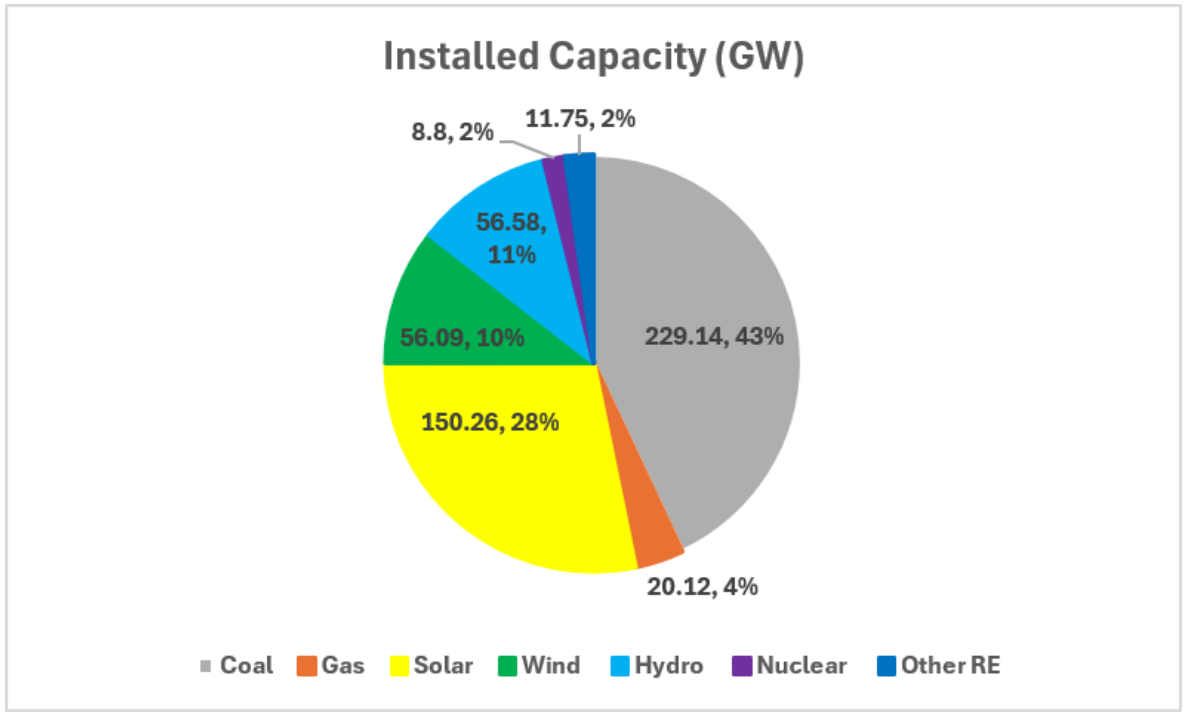


Accelerating Demand and Changing Supply Mix

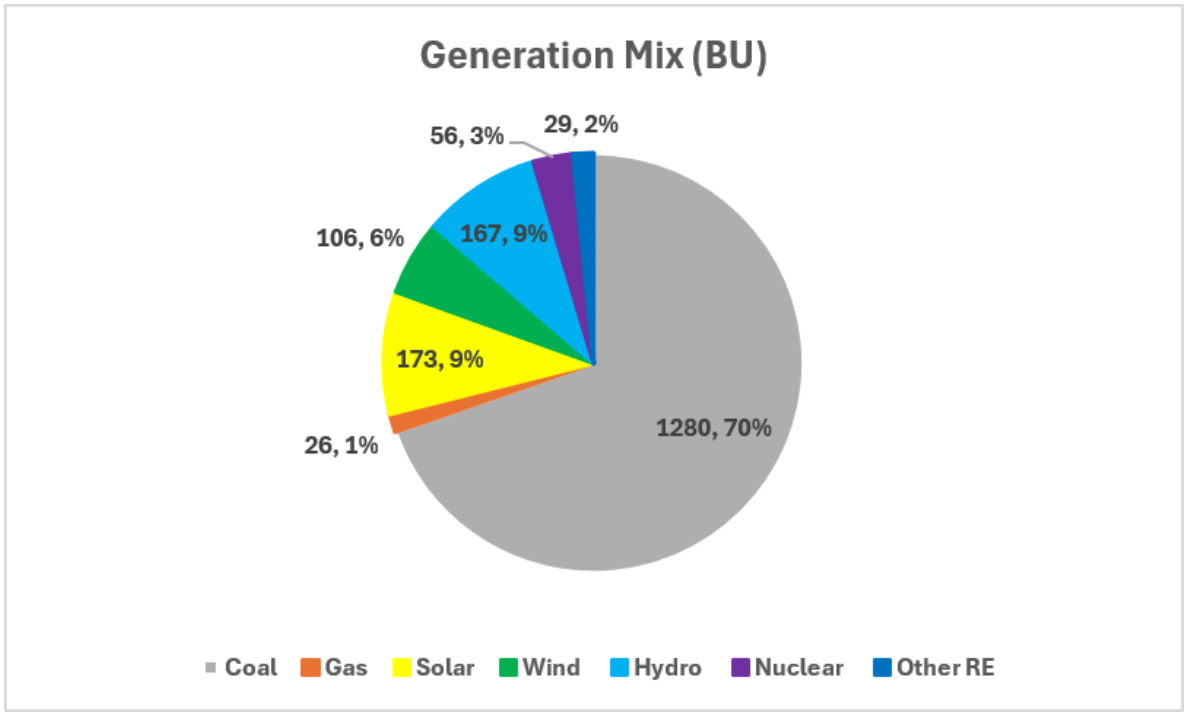
- **57% rise in peak demand (FY16–FY25)**
  156 GW → 245 GW
- **New peak demand met: ~270 GW (21 May 2026)**
  269 GW (Sept 2026) from 229 GW (Sept 2025): + 17% YoY
- **Supply mix for meeting 270 GW demand**
  Thermal 62.8% | Solar 22.0% | Hydro 5.8% | Wind 5.0%
- **Renewables are reshaping daytime generation**

Power Supply Position of India

Installed Capacity (In GW)



Generation Mix (In BU)



*For FY 2025-26

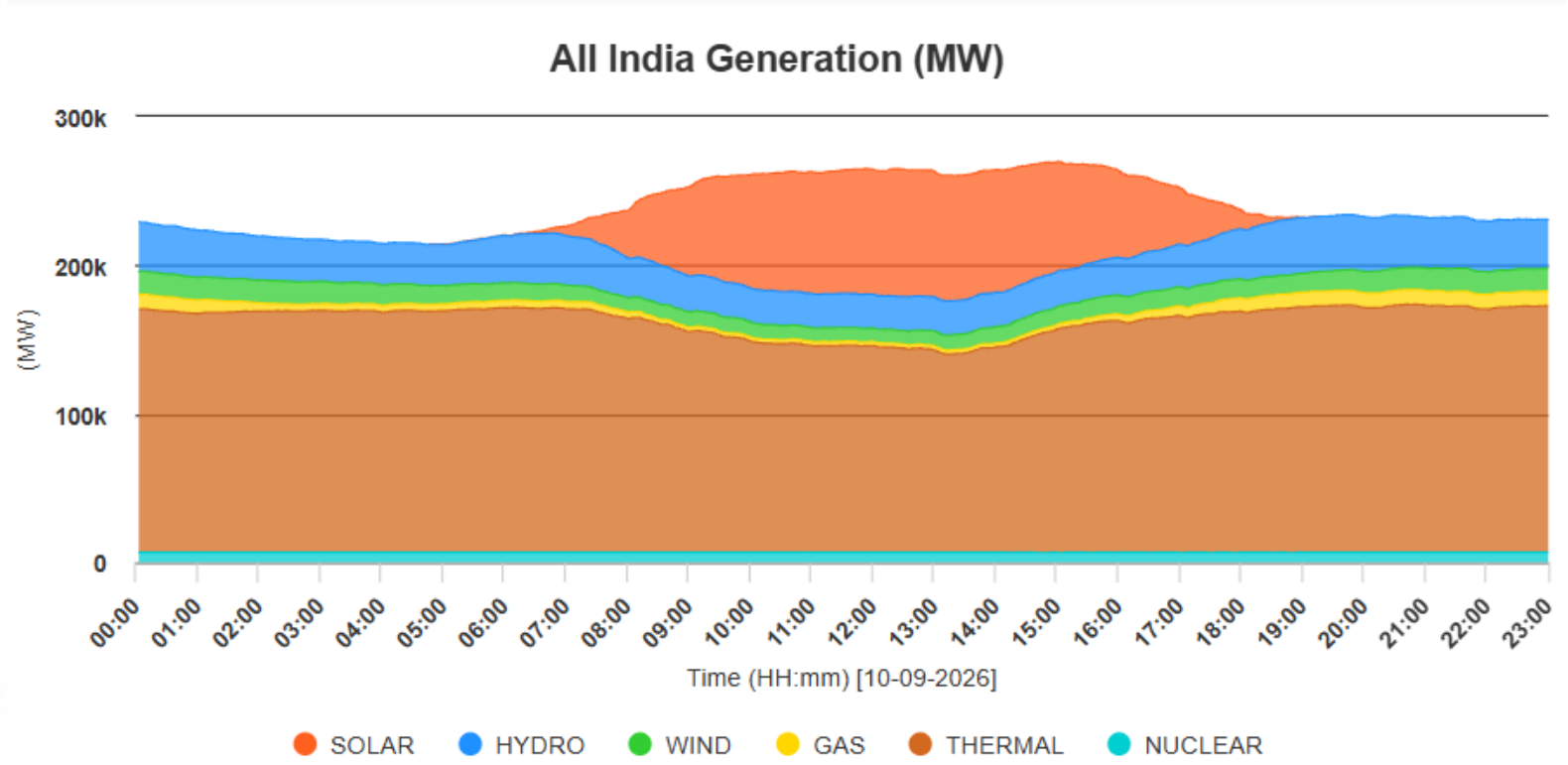
India’s Energy Storage Landscape

9.1 GWh
Operational BESS Capacity

12 GWh
BESS Capacity by Dec’26

7.2 GW
Operational PSP Capacity

94 GW PSP, 80 GW BESS
Required by 2035-36

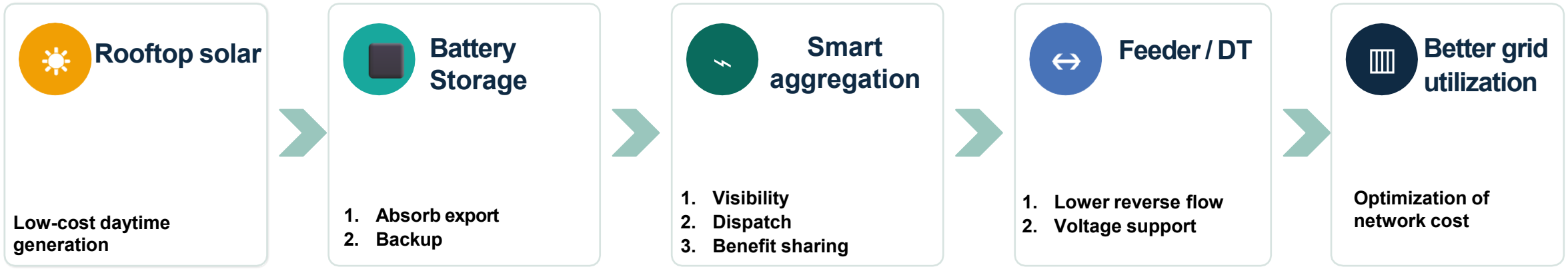


*Till Sept 2026

Dispatch stack of 10 Sept 2026

Imperative of BESS adoption at Decentralized Levels

DISCOMs gain greater feeder-level flexibility and enhance the value derived from each unit of solar generation, while consumers benefit from lower electricity bills.



CONSUMER VALUE

Lesser monthly bills • better backup

DISCOM / SYSTEM VALUE

Hosting capacity • peak shifting • asset utilisation

Pilot BESS Deployment in Rajasthan (RERC approved)

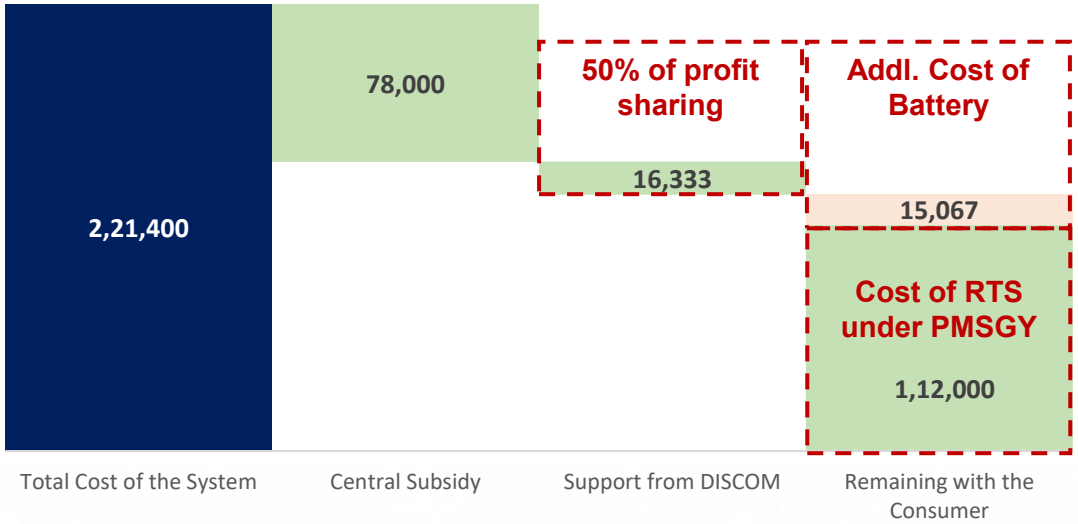
- **172 locations** identified for **1 MW/4 MWh standalone BESS** at **33/11 kV substations** with high RE penetration.
- **JVVNL:** 12 locations | **AVVNL:** 10 locations | **JDVVNL:** 150 locations.

South Bihar DISCOM | BTM Storage with RTS

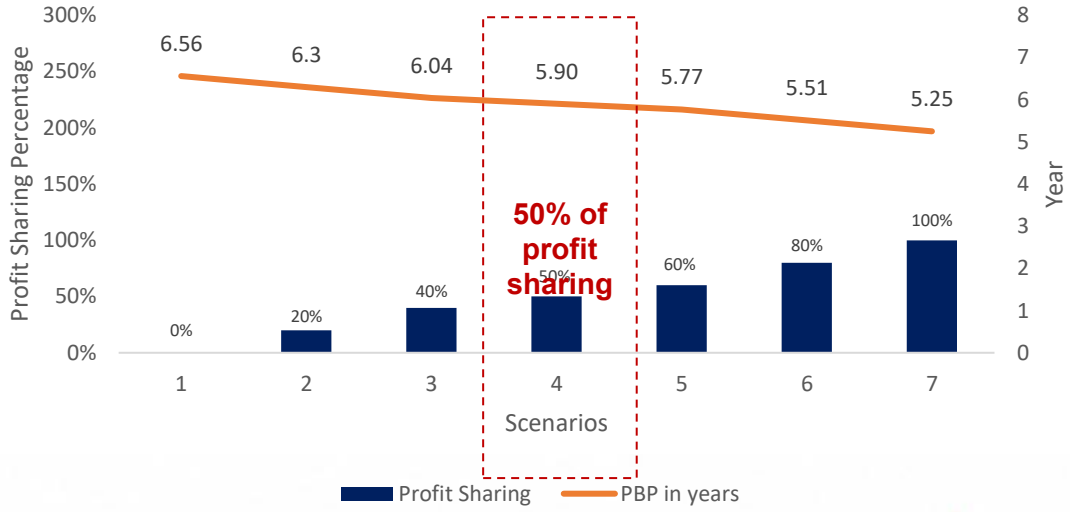
Context	Solution
▪ New rooftop capacity will increase daytime export and create network stress ▪ If this rollout is designed with BTM batteries from the outset, value can be captured systematically	▪ Consumer-side system ▪ Battery charges from surplus solar ▪ Battery reduces export / supports local consumption ▪ Utility benefits from avoided network usage and energy shifting

Scale opportunity	System Capacity	Calculation Basis
51, 494 households	3 kW solar + 4 kWh BESS	Export profile analysis of 57 consumers

Net Consumer Impact After Central, and DISCOM Support (proposed case)



Reduced Consumer Pay Back Period Through DISCOM Benefit Sharing with consumers

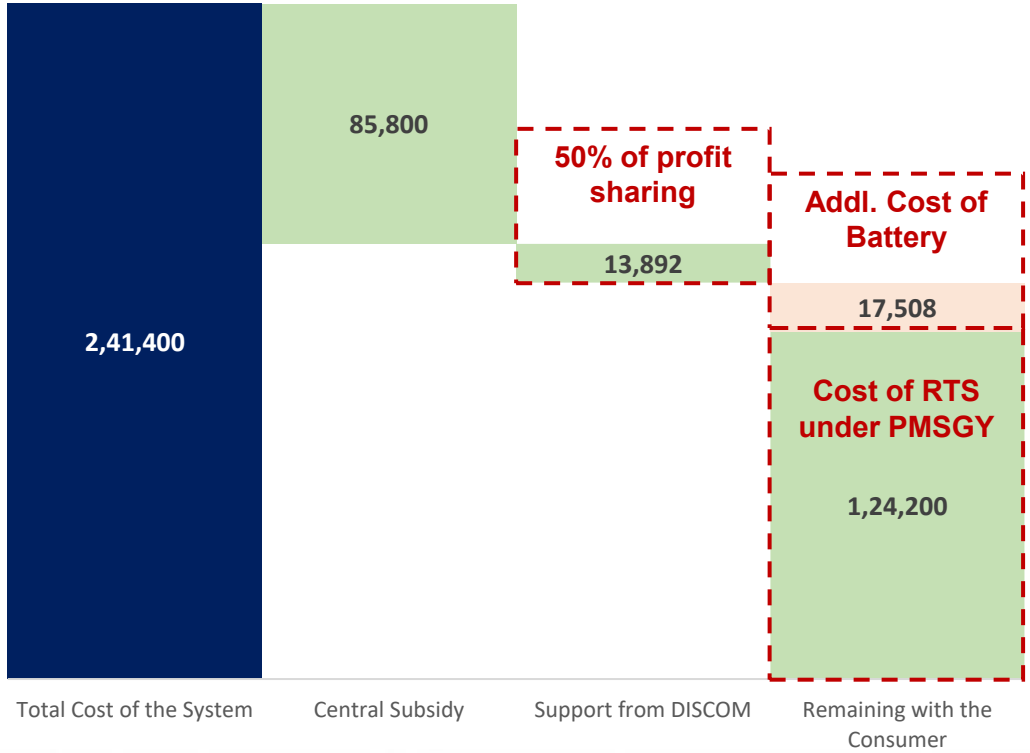


"DISCOM participation (through benefit-sharing) is a critical lever to unlock consumer adoption by improving payback and affordability"

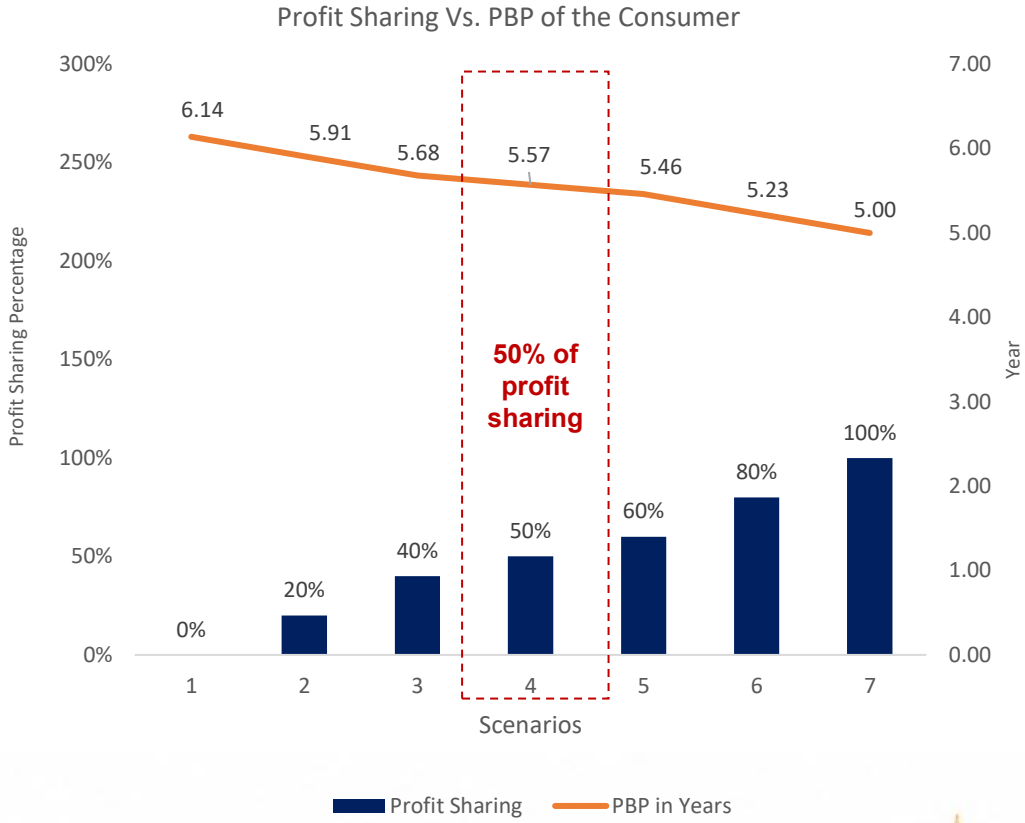
Assam DISCOM | BTM Storage with RTS

Scale opportunity	System Capacity	Calculation Basis
2.18 lakh households	3 kW solar + 4 kWh BESS	Export profile analysis of 175 consumers

Net Consumer Impact After Central and DISCOM Support (proposed case)



Reduced Consumer Pay Back Period Through DISCOM Benefit Sharing with consumers



"Sharing of partial DISCOM's realized benefits can materially improve consumer economics"

GUVNL, Gujrat | Grid-Scale BESS for Reverse Power Flow Mitigation

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Challenge

Increasing distributed RE penetration is creating operational challenges for GUVNL, including:

- Reverse power flow
- Voltage rise
- Inverter tripping
- Protection coordination issues

Proposed Solution

BESS for surplus solar absorption, reverse power flow mitigation, and energy arbitrage.

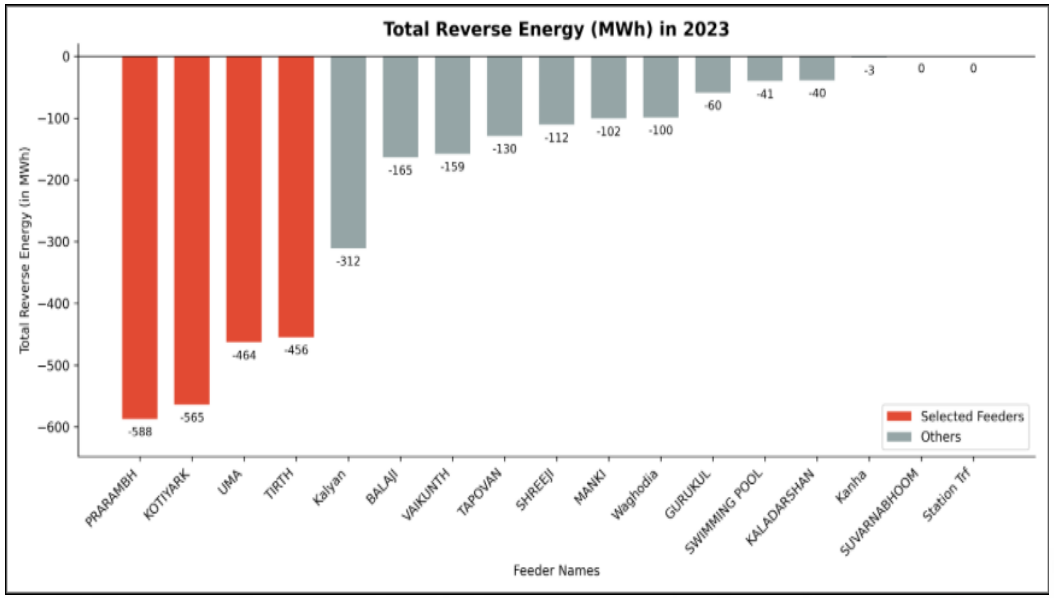
Objective

Evaluating optimal BESS sizing and siting for reverse power flow mitigation

Study Focus

Waghodia 66/11 kV Substation

Reverse energy flow: 3.29 GWh (2023) → 4.39 GWh (2030) | ~2% CAGR



GUVNL, Gujrat | Grid-Scale BESS for Reverse Power Flow Mitigation

Optimal BESS Capacity for RPF Mitigation

SIZING RESULTS – CAPACITY BUILT

Scenario 1	Scenario 2
Total Capacity **20 MW / 40 MWh**	Total Capacity **12 MW / 24 MWh**

Scenario 3	Scenario 4
Total Capacity **11 MW / 44 MWh**	Total Capacity **8 MW / 32 MWh**

Advantages of OPEX Model

- Technology provider bears performance, maintenance, and upgrade risks.
- Provides flexibility against rapid technology advancements and obsolescence.
- Pay-as-you-use structure aligns costs with operational benefits.
- Supported under MoP's BESS procurement guidelines (2022).

Challenges of CAPEX Model

- Requires high upfront capital investment and asset ownership.
- Limited flexibility due to fixed infrastructure investment.
- Higher risk of technology obsolescence and stranded assets.

Savings of around 11-30 Cr. were observed during the 4 scenarios in OPEX model

BRPL, NCT of Delhi | Kilokari & Shivalik BESS with GFM

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KILOKARI: Largest in South Asia – Grid Connected (Dist.) & India's First Commercially Approved BESS Project

TERI provided end-to-end transaction advisory to BSES Rajdhani Power Limited (BRPL) for a 20 MW/40 MWh standalone BESS project procured under Tariff-Based Competitive Bidding

1. **Tender Documentation**
   Drafting Scope of Work & Technical Specifications
2. **Eligibility & Evaluation Criteria**
   Framework for qualifying and scoring bidders
3. **BESS Agreement Structuring**
   Commercial & legal terms between utility and developer
4. **RfS Issuance & Bid Process Management**
   Managing the Request for Selection process
5. **Bid Evaluation & Reverse Auction**
   Technical & financial bid assessment, price discovery
6. **Regulatory Approval Support**
   Facilitating tariff adoption with DERC

Project Snapshot

Capacity	20 MW / 40 MWh (2 cycles/day)
	KBPL — Consortium of IndiGrid 2 Ltd & Amperehour Solar Technology Pvt Ltd
Off-taker	BSES Rajdhani Power Limited (BRPL)
Model	Build Own Operate Transfer (BOOT)
Concession Period	12 Years (8,760 cycles)
Concessional Funding	70% concessional debt from GEAPP (Global Energy Alliance for People & Planet)

Project Timeline

Feasibility
May – Jul 2023
Techno-commercial modelling, EOI, DERC in-principle approval

Procurement
Aug 2023 – May 2024
RFP, KBPL selection, BESS Agreement, tariff adoption

Commissioning
Jun 2024 – Apr 2025
Land handover, FAT/SAT, commissioning

Standalone BESS with GFM at Shivalik

PROJECT OVERVIEW

CAPACITY
12.5 MW / 25 MWh

BESS with
Grid Forming Inverter (GFM)

LOCATION
Shivalik, Delhi

STOREIN SUPPORT
Implementation support, technical analysis, RfP preparation & tendering support

FUNDING
Financial support for GFM Inverter

APPLICATIONS

- Energy Arbitrage / Energy Time-shifting
- Peak shaving
- Ancillary services
- Voltage support
- Reactive power support, black start, etc.

GRID-FORMING INVERTER (GFM)

Capable of operating in grid-forming mode, allowing it to:

- ✓ Establish voltage and frequency reference
- ✓ Energize feeders
- ✓ Support grid restoration

This feature enables the battery system to act as a virtual generator in the network.

BESS Sizing for APEPDCL (BESS with PM-KUSUM)

Scope of study: Assessment of how adding PV changes battery size, reverse power flow (RPF) and arbitrage economics at two substations with contrasting load-to-solar ratios.

Nimmada S/S carries nearly twice **Rajapuram S/S's** load, is flat across seasons and peaks in the daytime, which makes it a natural sink for solar.

What we assessed

1. **Battery sizing**
   Minimum energy (MWh) and power (MW) needed to keep export near zero

2. **RPF reduction**
   Share of reverse power flow suppressed by the battery

3. **Arbitrage economics**
   Capital cost and impact on the end consumer (ARR) under the CAPEX and OPEX business models

Dispatch priority: Prioritise zero export, with residual battery capacity deployed for IEX RTM arbitrage

Scenarios modelled

	Nimmada SS	Rajapuram SS
Base case	2.5 MW PV	4.14 MW PV
Scenario 2	3.5 MW PV (+1 MW)	6.14 MW PV (+2 MW)
Scenario 3	4.5 MW PV (+2 MW)	8.14 MW PV (+4 MW)
Load-to-solar ratio (base case)	3.74	1.42
RPF suppression	*Target: ~98% in all scenarios*	

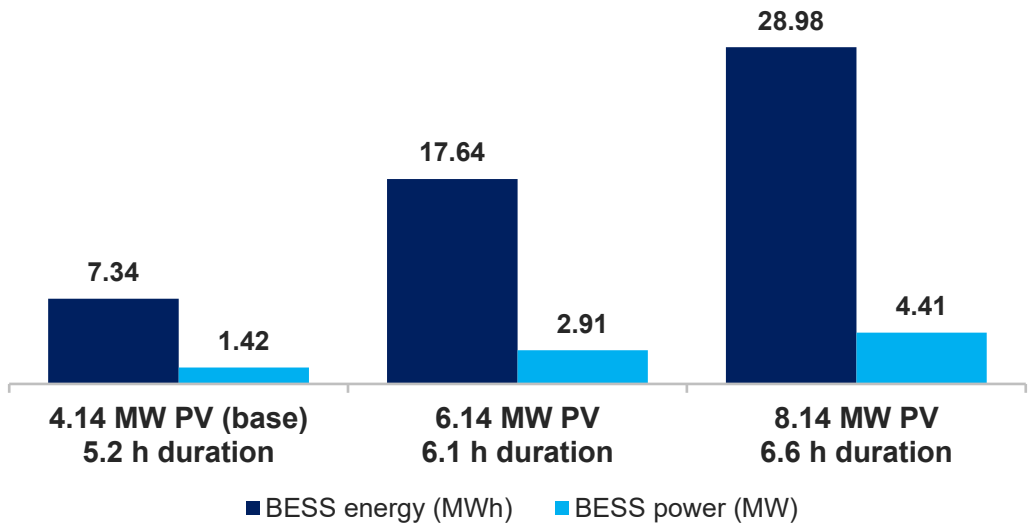
Rajapuram must shift midday solar to its evening peak; Nimmada absorbs most solar directly

BESS Sizing for APEPDCL (BESS with PM-KUSUM) | Findings

Rajapuram S/S

Each extra MW of solar needs ~5.4 MWh and ~0.75 MW of BESS; ARR rises under CAPEX but stays nearly flat under OPEX

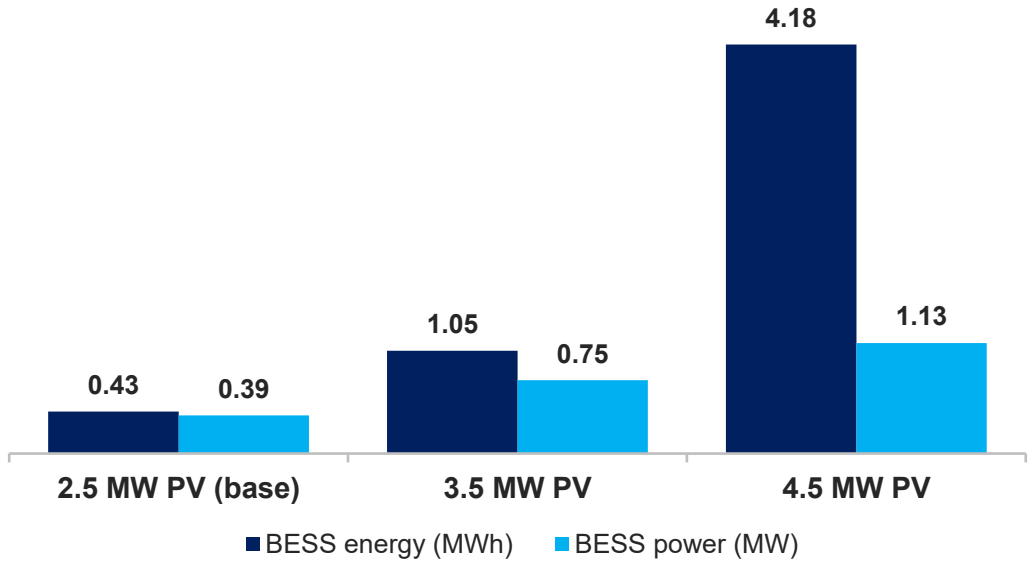
BESS capacity required in each PV case



Nimmada S/S

BESS power grows steadily with solar, but energy needs jump beyond 3.5 MW; the best PV size differs by business model

BESS capacity required in each PV case



OPEX gives the larger ARR reduction (up to ₹22.2 Cr NPV at Nimmada 4.5 MW). Under CAPEX, Rajapuram's ARR rises beyond the base case (+₹28.65 Cr NPV at 8.14 MW).c

TERI's Work in BESS

COMPLETED PROJECTS

- **UI-ASSIST (US-India Collaborative for Smart Distribution System with Storage):** Techno-economic feasibility studies for distribution-level pilots, and study on role of DSOs in India.
- **Technical study** on BESS sizing/siting and operations for selected distribution feeders of WBSEDCL
- A study to assess the role of **urban micro-grids in strengthening distribution networks**, especially with rising renewable energy penetration.
- **Mobile Substation and Grid Storage System (MOBISUB)** Pilot for providing supply during substation maintenance & peak demand in distribution network.
- **20 MW/40 MWh BESS, BRPL-Delhi:** Bid process management under Tariff-Based Competitive Bidding — India's first commercial standalone distribution-level BESS project.
- Supported KfW for **identifications of grid scale BESS projects for financing & technical assistance** in India
- **Detailed Project Report preparation for BESS integration** (for CESC & GRIDCO)
- **Digital Twin for distribution grid operation:** BESS dispatch optimization, Automated Demand Response and DT health monitoring pilot projects.

ONGOING PROJECTS

- **Feasibility studies (techno-economic analysis) for BESS Project Implementation** for Discom (GUVNL, APEPDCL)
- **AgriPV with BESS** Techno-economic studies (BRPL).
- **Behind-the-Meter (BTM) BESS** Techno-economic analysis (APDCL, SBPDCL).
- **RfS preparation & tendering support** for distribution-level BESS (BRPL, BYPL).
- **Capacity-building programs** on BESS for DISCOMs, Banking/Financing institutions, ITIs & OEMs
  (Conducted training programs for BRPL, BYPL, GUVNL, AP Power Utilities)

UPCOMING PROJECTS

- GRIDCO: BESS Potential analysis for 250 sub-stations
- TSECL: Feasibility study for BESS integration at 10 sub-stations study on establishment of EVCI integrated with BESS in UT of Ladakh
- Assessment of India's BESS market to identify business opportunities, partnerships, and market entry strategies
- APEPDCL Grid Readiness Study with DRE integration AgriPV + BESS technical study for MP

THANK YOU

For discussions/suggestions/queries email:
dum@indiasmartgrid.org
www.dumindia.in



Alekhya Datta
Director, Electricity & Renewables (TERI)
alekhya.datta@teri.res.in
+91-9999466210
https://www.teriin.org/profile/alekhya-datta

TERI's Energy Programme